

HYDROCARBON

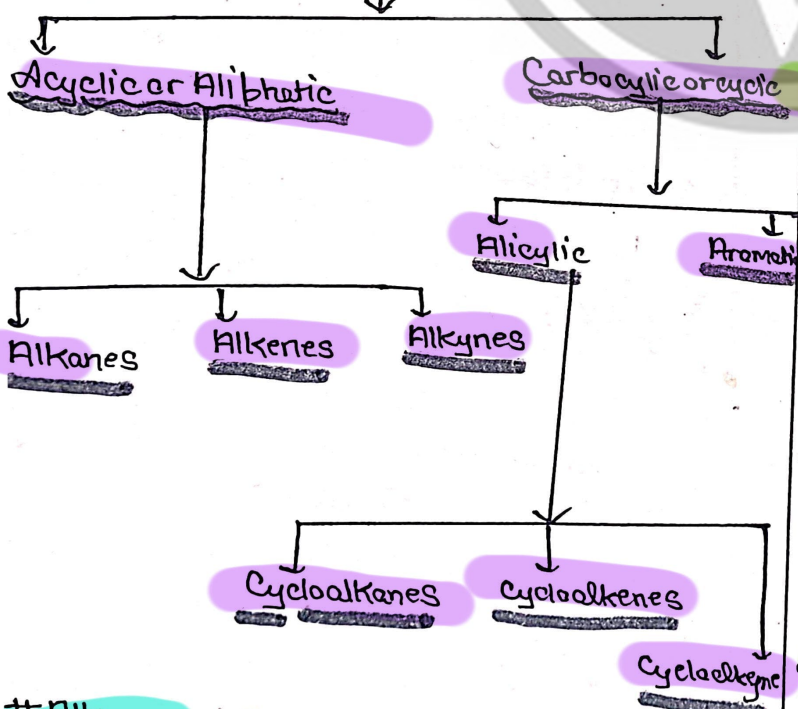
Hydrocarbons are composed of Carbon and Hydrogen.

The important fuel like petrol, kerosene, coal gas, CNG, LPG, etc. are all hydrocarbons or their mixture.

Sources: Petroleum and natural gas are the major sources of aliphatic hydrocarbon while coal is an important source of aromatic hydrocarbons. The oil trapped in the rock is known as Petroleum. PETRA-ROCK, OLEUM-OIL, The oil in the petroleum field is covered with a gaseous mixture known as natural gas. The main constituents of the natural gas are methane, ethane, propane and butane.

CLASSIFICATION OF HYDROCARBONS:

Hydrocarbon



Alkanes:

- Paraffins.
- General Formula C_nH_{2n+2}

sp^3 hybridisation.

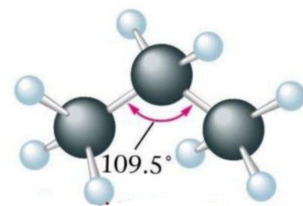
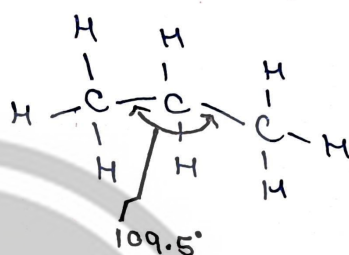
C-C bond length $1.54 \text{ \AA}$

Chemically unreactive

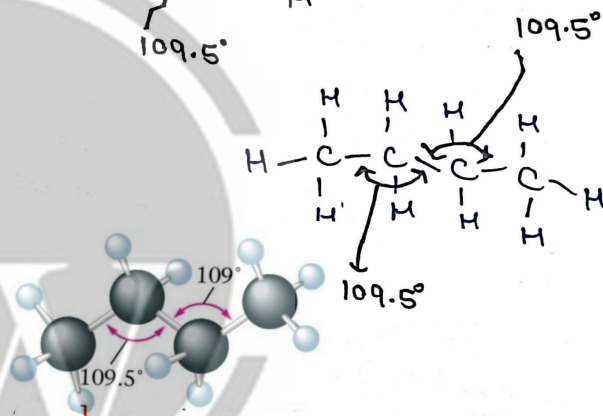
Show chain, position and optical isomerism

Heptane has 9 isomer, Octane 18 and decane 45.

Nomenclature:

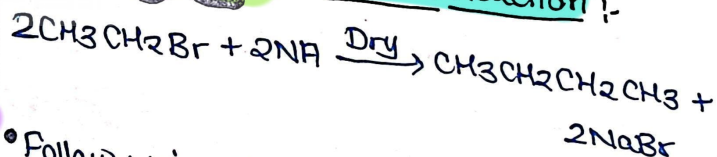


(a) Propane



(b) Butane

Preparation: i) Wurtz Reaction:-

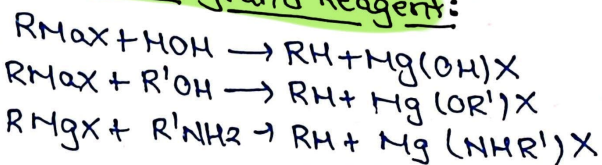


- Follow mainly free radical mechanism.
- Useful in preparing an alkane containing even number of carbon atoms.
- Stepping up reaction.

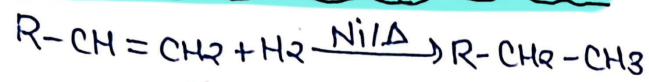
ii) Frankland-Reaction:-



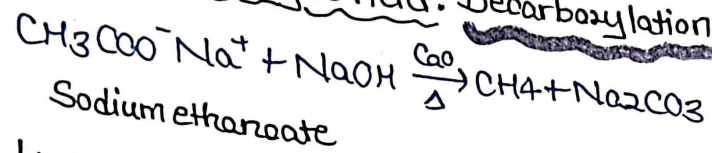
From Grignard Reagent:



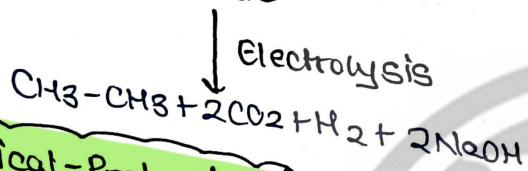
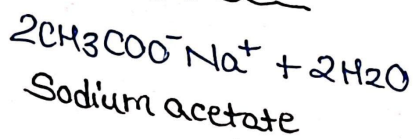
iii) From unsaturated hydrocarbons:



iv) From Carboxylic Acid: Decarboxylation-



Kolbe's electrolytic Method-



Physical-Properties:

i) Nature: Non-polar due to covalent nature of C-C bond and C-H bond. C-C bond energy = 83 kJ/mole and C-H bond energy = 99 kJ/mole.

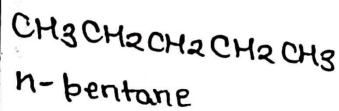
ii) Solubility: Like dissolve like. Viz, polar compounds dissolve in polar-solvent and Non-polar compound dissolve in non-polar solvent.

iii) Boiling-Point: Low boiling point due to non-polar in nature.

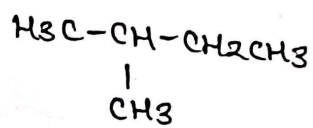
- The molecules are held together only by weak van der Waals' forces.
- Since we know that the magnitude of van der Waals' forces is directly proportion to molecular size. Therefore, the boiling point increase with increase the molecular size i.e. with increase in number of carbon atoms.

Noted: The boiling points of the branched chain alkanes are less than the straight chain isomers.

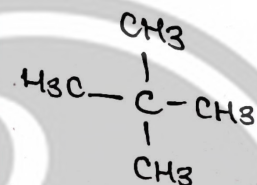
• This is due to the fact that branching of the chain makes the molecule more compact and thereby decrease the surface area and consequently, the magnitudes of Vander Wall's forces also decreases.



(boiling point = 309K)



Iso-pentane
boiling point = 301K

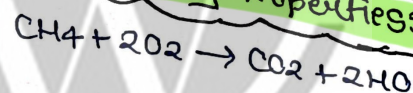


neo-pentane

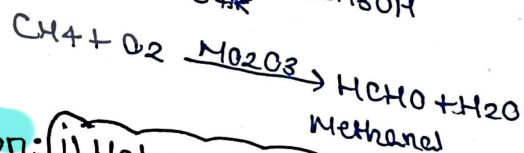
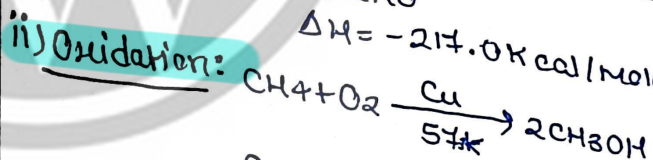
boiling-point = 282.5K

Chemically-Properties:

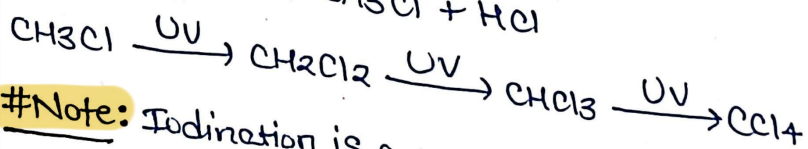
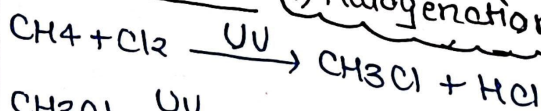
i) Combustion:



$\Delta H = -217.0 \text{ K cal/mole}$



iii) Substitution: (i) Halogenation:

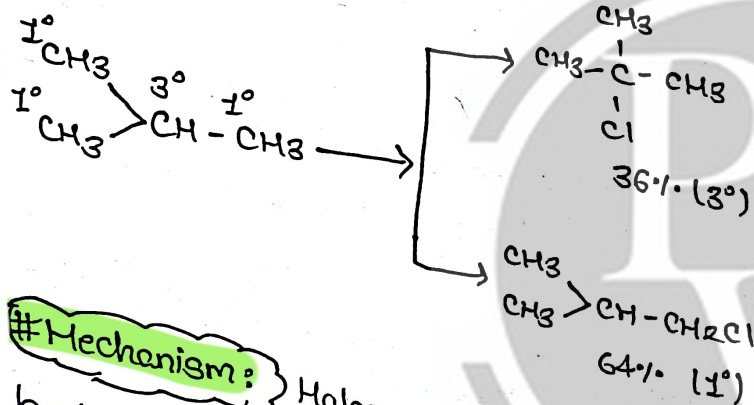
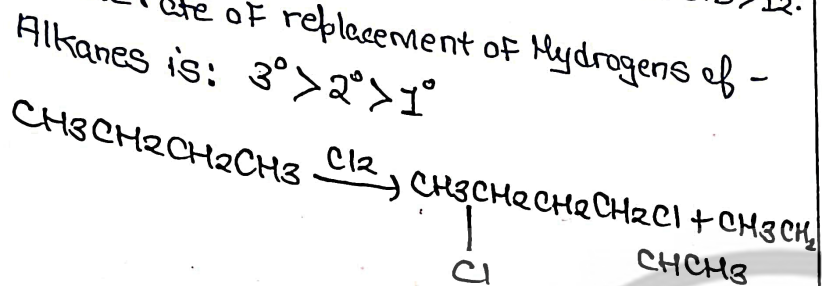


Note: Iodination is a reversible rxn. So it is carried out by heating alkane in the presence of some oxidising agent like Iodic Acid (HIO₃) or nitric acid (HNO₃) or mercuric oxide (HgO) which oxidizes HI formed during the reaction.



Features of Halogenations:

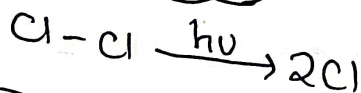
- i) The reactivity of Halogens: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.
- ii) The rate of replacement of Hydrogens of Alkanes is: $3^\circ > 2^\circ > 1^\circ$



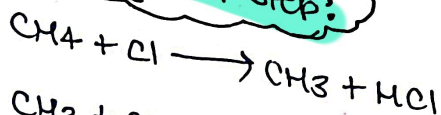
Mechanism:

Halogenations rxn take place by free radical mechanism. The reaction proceed into the following steps.

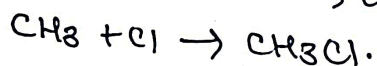
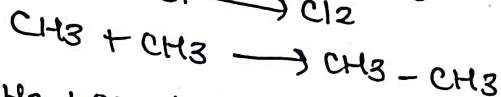
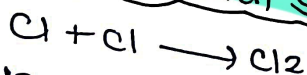
i) Chain Initiation Step:



ii) Chain Propagation Step:

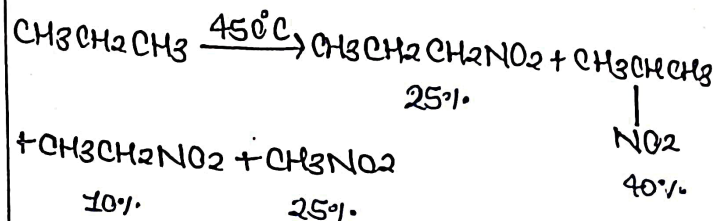


iii) Chain termination Step:

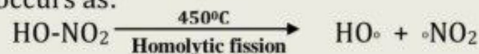


(ii) Nitration: The rxn takes place by free radicals mechanism at high temp (450°C).

At high temp C-C bond is also broken so that mixture of nitroalkanes is obtained.

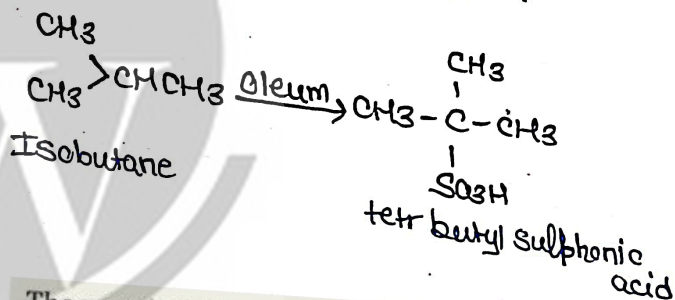


The reaction occurs as:

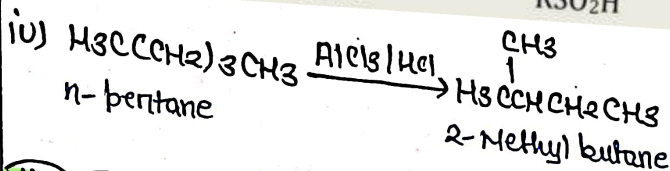
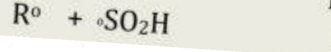
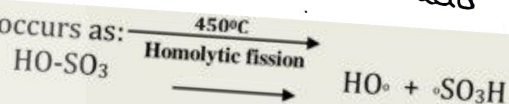


iii) Sulphonations:

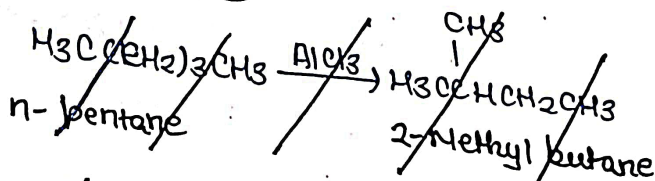
Replacement of hydrogen atom of alkane by $-\text{SO}_3\text{H}$ group.



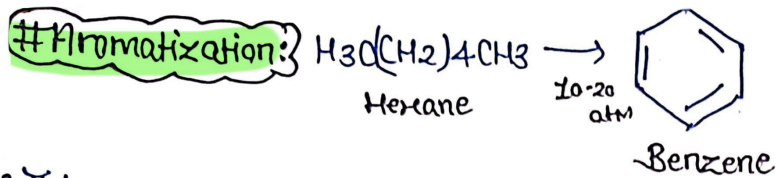
The reaction occurs as:



v) Aromatization:

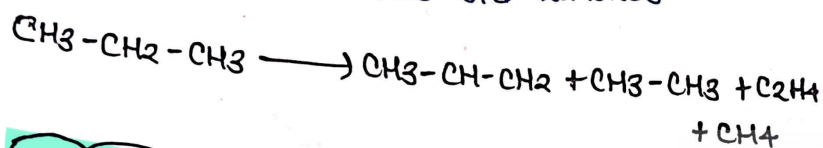


This method also called dehydrogenation or hydroforming.

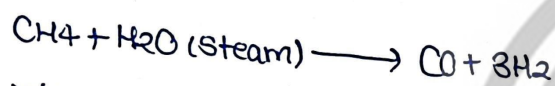


• This method is also called dehydrogenation or hydroforming.

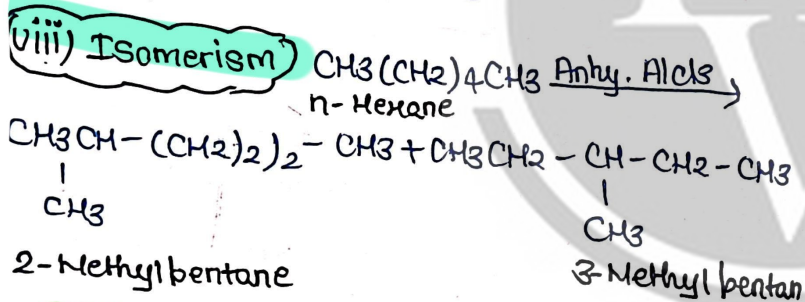
(vi) Thermal-decomposition: When higher alkane are heated at high temp (about 700-800K) in the presence of alumina, the alkanes break down the lower alkanes and alkenes.



(vii) Action of Steam: catalyst: Nickel, alumina Al_2O_3

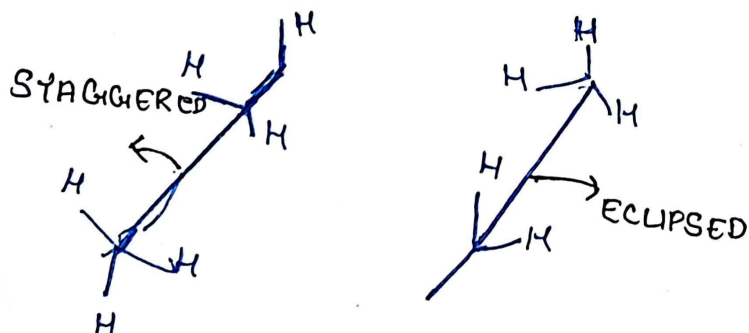


This reaction is used for the industrial preparation of hydrogen from natural gas.

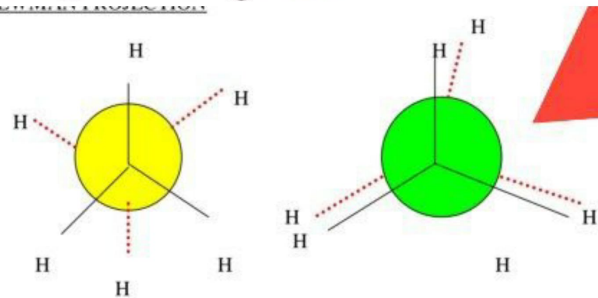


#Conformational Isomerism: The different molecular arrangements arising as a result of rotation around carbon carbon single bond are called Conformational Isomerism.

#Sawhorse Representation

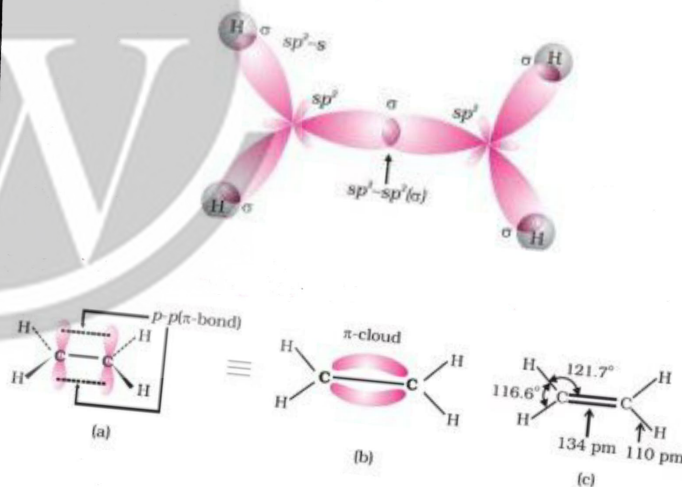


#Newman Projection

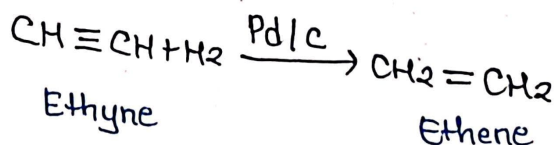


#Alkenes: Unsaturated hydrocarbon - which have double bond.

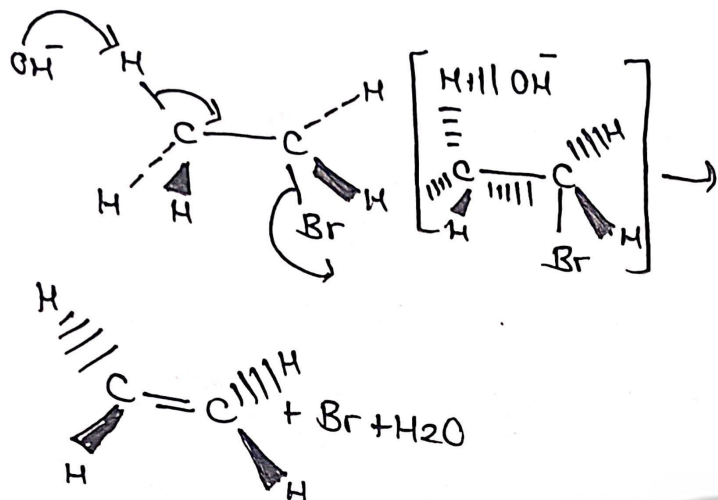
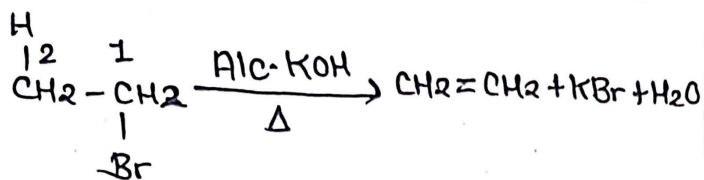
- General molecular formula C_nH_{2n} .
- C-C bond hybridization $\pm 34^\circ$.
- sp^2 hybridization
- When we treated alkane with chlorine, oily products are obtained. So Alkenes are called Olefins.
- Structure of Double bond:-



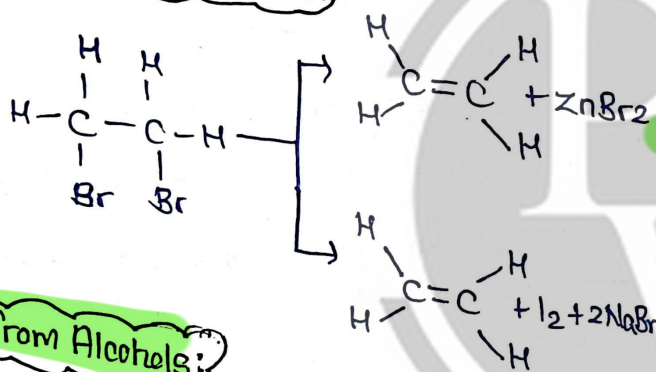
#Preparation: i) **From Alkynes:** Alkynes on partial reduction with partially deactivated palladium charcoal known as Lindlar's - catalyst give alkynes.



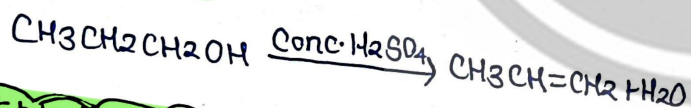
ii) **From Haloalkanes** - Dehydrohalogenation



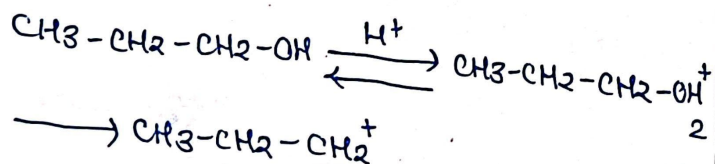
From dihaloalkanes:



From Alcohols:

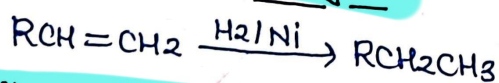


Mechanism:

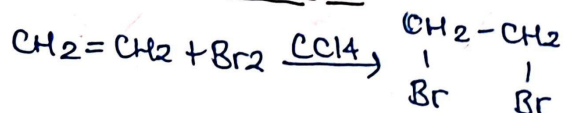


Chemical Properties: Addition-Rxn:

i) Addition of Hydrogen:



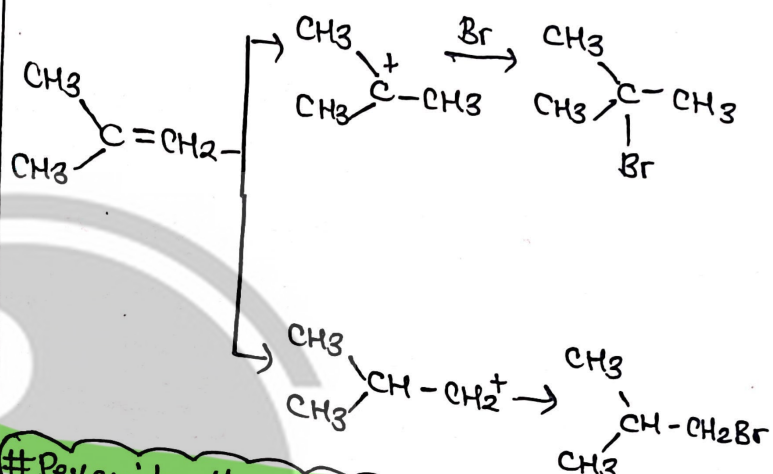
ii) Addition of Halogens:



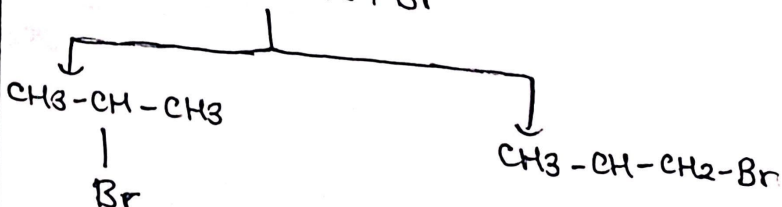
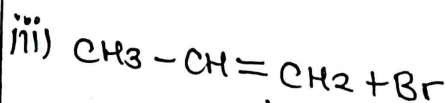
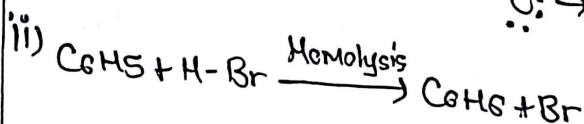
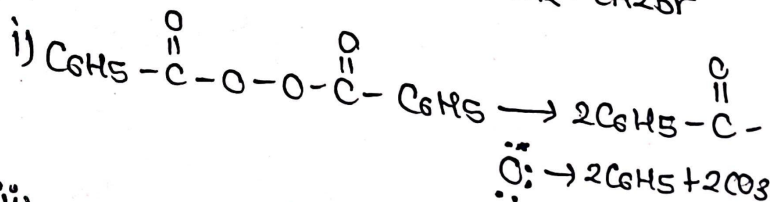
iii) Addition of Hydrogen halides:

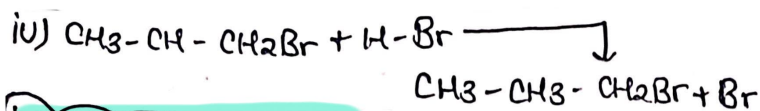


Markovnikov-Rule: Negative part of the addendum gets attached to that carbon atom which possesses lesser number of hydrogen atom.

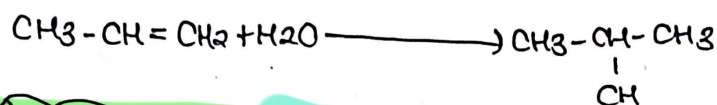


Peroxide effect of Kharasch: In 1933 Kharasch and Mayo observed that when HBr is added to an unsymmetrical double bond in the presence of organic peroxide the rxn takes place opposite to the Markovnikov rule.



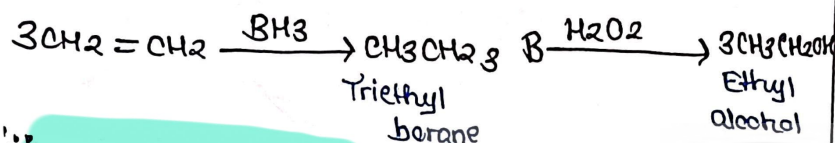


iv) Addition of water: Acid catalyzed addition of water.

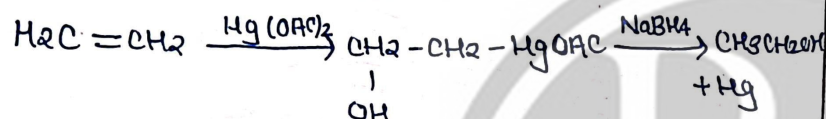


2) Oxidation: i) **Combustion** $\rightarrow \text{CO}_2 + \text{H}_2\text{O}$

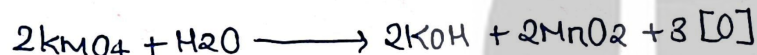
i) **Hydroboration-oxidation:** Alkenes react with diborane to form trialkyl boranes which on oxidation with alkaline H_2O_2 give alcohols.



ii) **Oxymercuration - demercuration:**

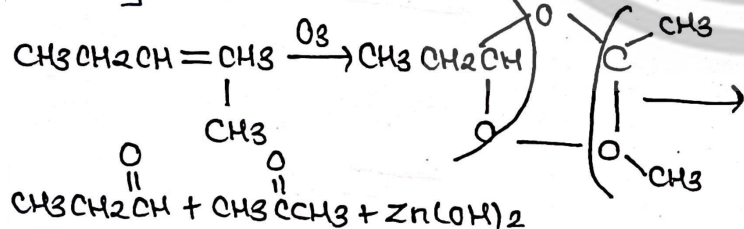


iv) **Oxidation with Potassium permanganate:**



- This rxn also called hydroxylation.
- Cis product i.e. cis-1,2-diol is obtained.

Oxidation with Ozone: Ozonolysis gives carbonyl compounds.

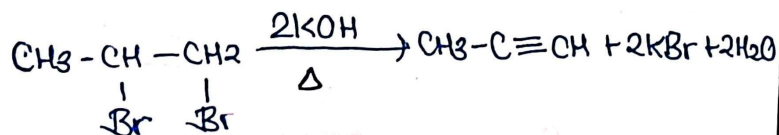


Alkynes: $\rightarrow$ Unsaturated hydrocarbon which have triple bond

$\rightarrow$ General molecular formula.

$\rightarrow$ SP hybridization.

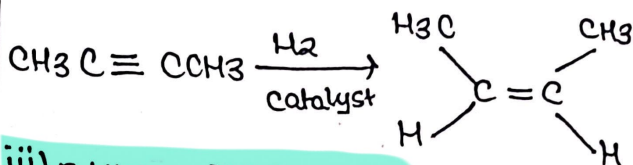
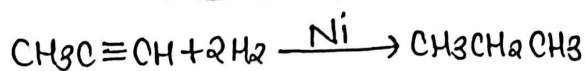
Preparation: From vicinal dihalides: - dihalogenation



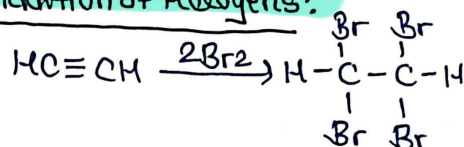
Chemical Properties:

i) **Addition Reaction** - Alkyne show electrophilic addition reaction.

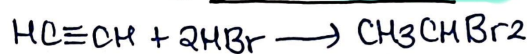
ii) **Addition of Hydrogen** - Hydrogenation.



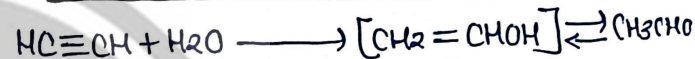
iii) **Addition of Halogens:**



iv) **Addition of hydrogen halides:**



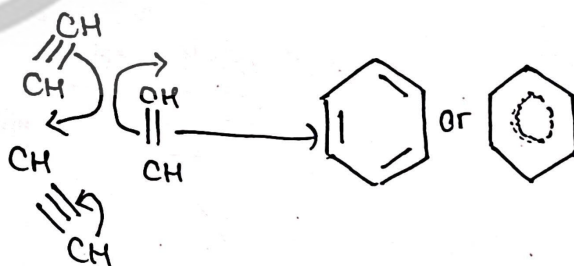
v) **Addition of water (Hydration):**



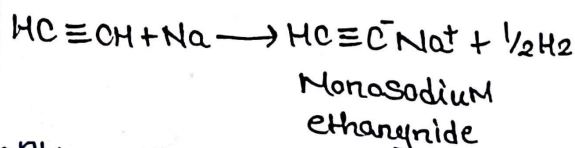
(5) Polymersitation: a) **Linear Polymerisation:**

of ethyne gives Polyacetylene or polyethyne which is a high molecular weight polyene containing repeating unit of $(\text{CH} = \text{CH} - \text{CH} = \text{CH})$ and can be represented as $-(\text{CH} = \text{CH} - \text{CH} = \text{CH})_n$

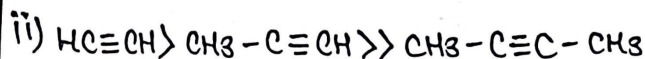
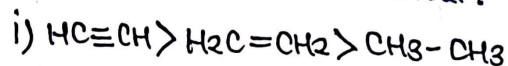
b) **Cyclic Polymerisation:** Result in the form of aromatic compound.



Acidity of Alkynes: Terminal alkynes are acidic in nature.

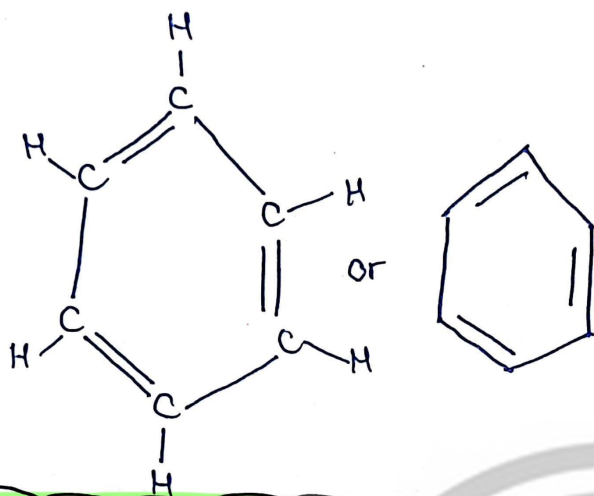


• Alkanes, Alkenes, alkynes Follow the following trend in their acidic behaviour:

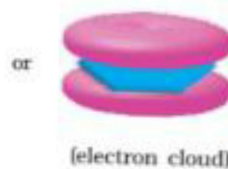
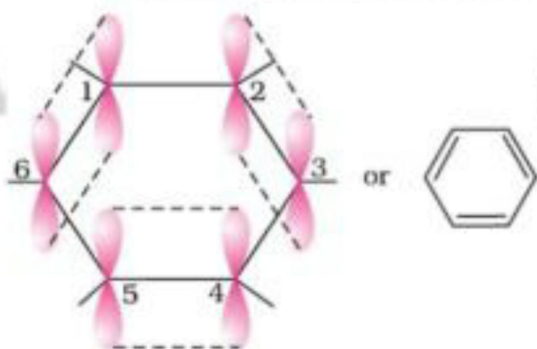
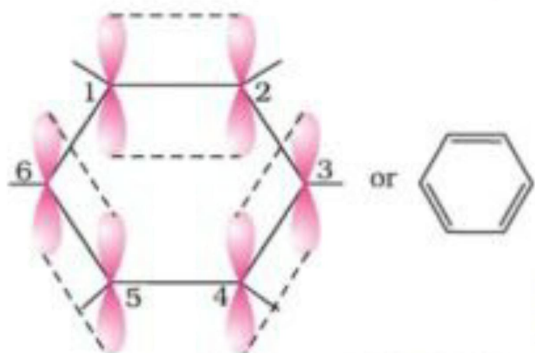
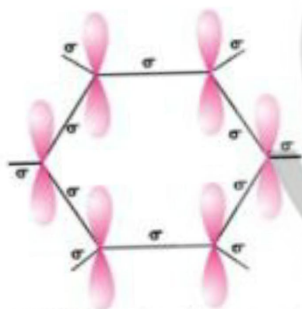


Aromatic-Hydrocarbon: Aromatic compound containing benzene ring are known as benzenoids and those not containing a benzene ring are known as non-benzenoids.

Structure of Benzene:



Resonance and Stability of Benzene: All the six carbon atoms in benzene are sp^2 hybridized and these hybrid orbitals form sigma bonds.

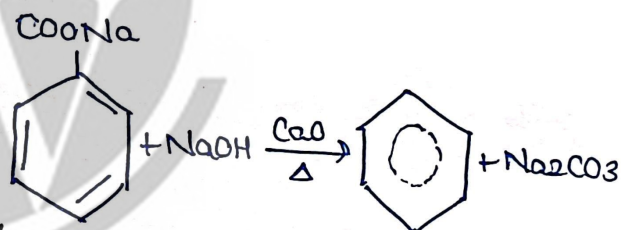


Aromaticity: The compounds that follow the following features are to be considered aromatic.

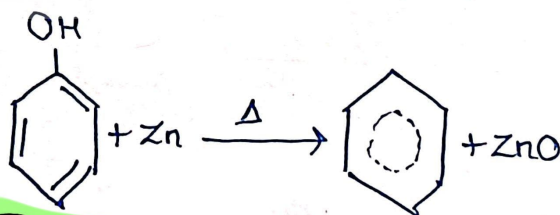
- Planarity.
- Complete delocalisation of π electron in the ring.
- Presence of $(4n+2)$ π electron in the ring where n is an integer ($n=0,1,2,\dots$). This is often referred to as Hückel Rule.

Preparation of Benzene: i) **Cyclic Polymerisation of ethyne:**

ii) **Decarboxylation of Aromatic acids:**



iii) **Reduction of Phenol:**

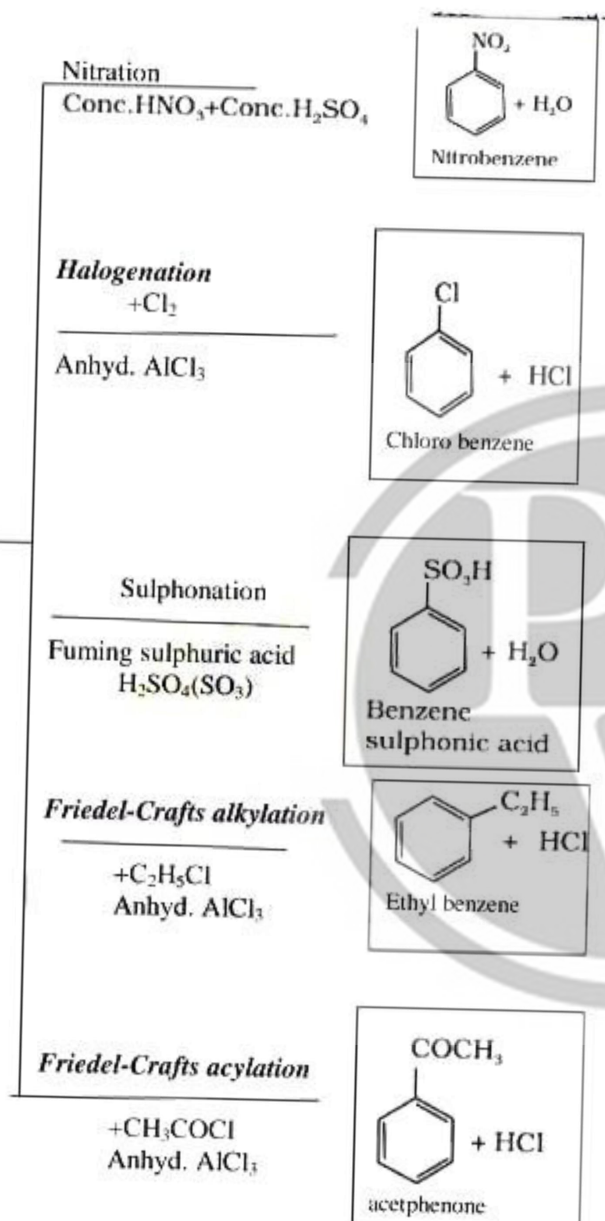


Physical-Properties:

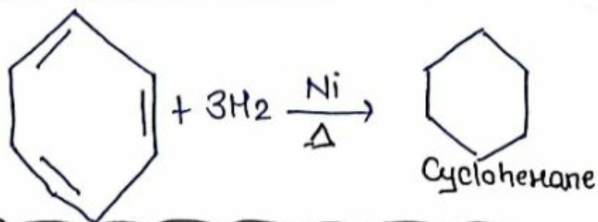
- Aromatic hydrocarbons are non-polar molecules and are usually colourless liquids with characteristic aroma.
- Aromatic hydrocarbons are immiscible with water but are readily miscible with organic solvents.
- They burn with sooty flame.

Chemical - Properties:

- Generation of the electrophile.
- Formation of carbocation intermediate.
- Removal of Proton From the carbocation intermediate.

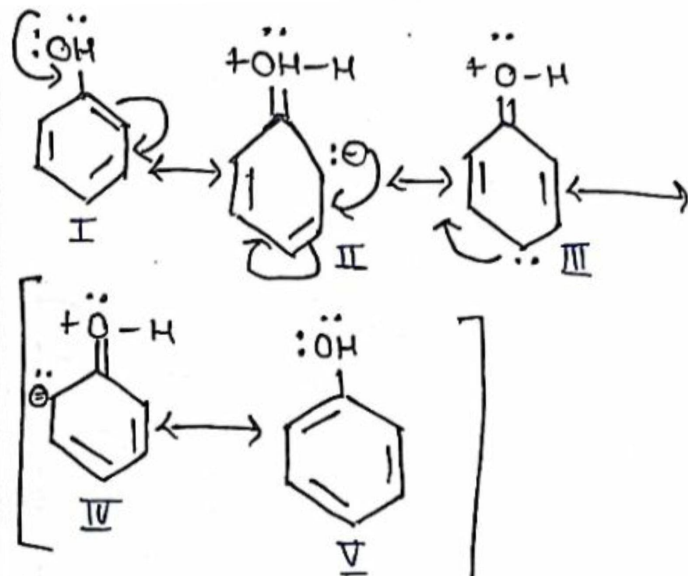


Addition of Benzene:



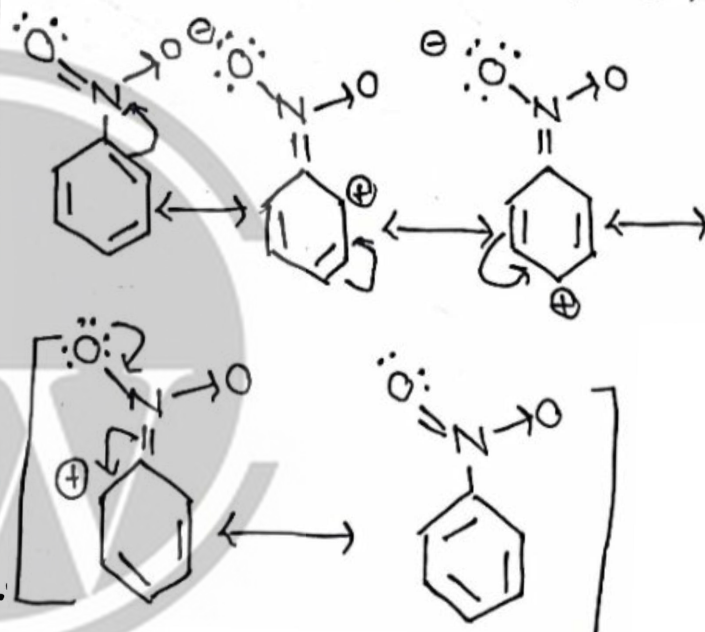
Directive influence of a functional group in monosubstituted benzene:

- Ortho and para directing group and activating**
 $-\text{OH}, -\text{NH}_2, -\text{NHR}, -\text{NHCOCH}_3, -\text{OCH}_3, -\text{CH}_3, -\text{C}_2\text{H}_5$, etc.



ii) Meta directing group of deactivating

$-\text{NO}_2, -\text{CN}, -\text{CHO}, -\text{COR}, -\text{COOH}, -\text{COOR}, -\text{SO}_3\text{H}$, etc



iii) Ortho and para directing group of deactivating:

Halogens because of their strong $-\text{I}$ effect, overall, electron density on benzene is decreasing.

CARCINOGENICITY AND TOXICITY: Benzene and polynuclear hydrocarbons containing more than two benzene rings fused together are toxic and said to possess cancer producing property.